



OPERATIONAL TECHNOLOGY (OT) CYBERSECURITY

WHAT IS OT?

Operational Technology (OT) is a specific category of systems that directly control physical equipment or monitor conditions that cause equipment to move or respond.

OT differs from IT (email, office computers) because OT involves physical equipment, processes, and safety implications.

These all meet the definition because they interact with or directly manage the physical environment.

Simple rule: If a system starts, stops, opens, closes, moves, alarms, measures, or controls equipment, it is OT.

Examples of common OT systems found at facilities and vessels:

- Pumps, valves, motors, actuators (devices that cause movement or change)
- PLCs (Programmable Logic Controllers) that issue commands to equipment
- HMIs (Human-Machine Interfaces) or operator screens used to monitor and control operations
- SCADA systems that collect data and control processes remotely
- Alarm and safety shutdown systems that detect conditions and trigger responses
- Power distribution and automated breaker systems
- Cargo handling, transfer, and loading automation
- Navigation or propulsion control systems that control or influence vessel movement (AIS, ECDIS, autopilot)

COMMON OT SYSTEMS YOU MAY USE

Below are the most commonly used OT systems at our facilities and aboard vessels in the fleet.

FACILITIES	VESSELS
CCTV cameras and video management systems	CCTV cameras and video management systems
Gate or building access control systems (card readers, badge readers)	Vessel automation systems
Fire alarm and sprinkler systems	Engine control systems (ECS)
Interior lighting systems	Electronic Chart Display and Information Systems (ECDIS)
License plate readers	Integrated Bridge Systems (IBS)
	Navigation and positioning systems (AIS)
	Ballast water management systems (BWMS)
	Programmable logic controllers (PLCs)
	Global positioning systems (GPS)
	Voyage data recorders (VDR)
	Fire detection and suppression controls



WHY OT CYBERSECURITY MATTERS

Cyberattacks aren't just about stealing data. When OT systems are affected, the impact can be physical, not just digital.

Cyber issues in OT can lead to:	What this can look like in real life:
• Equipment operating in unexpected or unsafe ways • Loss of visibility or control • Safety or environmental hazards • Operational disruption	• A pump turning on or off unexpectedly • A valve opening or closing without a command • Alarms or operator screens showing incorrect information • Sensors providing inconsistent or impossible values • Propulsion, steering, or thruster responding differently than commanded • Navigation or bridge systems (AIS, ECDIS, autopilot) showing conflicting or unreliable data

COMMON OT CYBER RISKS

Most OT problems start with everyday actions, not hacking. If something feels like it "cuts corners," it may create risk. Avoid these risky behaviors:

- Sharing logins or using someone else's credentials
- Plugging in personal or unknown USB devices
- Letting vendors connect equipment without approval
- Using unapproved tools or laptops
- Propping open secure OT areas
- Changing setpoints outside approved procedures

WARNING SIGNS SOMETHING MAY BE WRONG

Report any of the following immediately:

- Screens freeze, flicker, or show odd readings
- Equipment behaves differently than commanded
- Alarms don't match what's happening in the field
- Unknown cables, USBs, or devices appear
- Setpoints change on their own
- A vendor asks to connect unfamiliar equipment
- Logins suddenly stop working

WHAT TO DO IF YOU NOTICE SOMETHING

If you see something unusual, report it right away. You will never get in trouble for reporting a concern that turns out to be nothing.

Report to:

1. Supervisor / Operations Lead
2. Facility / Vessel / Company Security Officer (FSO / VSO / CSO)
3. IT ServiceDesk
4. Cybersecurity Officer (CySO)

**IF YOU SEE
SOMETHING, SAY
SOMETHING!**